



WIPRO NGA SELENIUM WITH PYTHON

APRIL 2026 | BATCH 2

CAPSTONE PROJECT

PROJECT CASE STUDY – E-COMMERCE

PROJECT WEBSITE – NYKAA

DATE: 25-05-2026

CONTRIBUTION OF THE TEAM

- Yoga Lakshmi N – Men Category
- Kunal Chauhan – Fragrance Category
- Nishant Kumar – Hair Category
- Alish Rani – Makeup Category
- Harsh Singh – Skin Category
- Krishna Kant Soni – Health and Wellness Category
- Badrinadh Cheetirala – Mom and Baby Category

Test Automation Overview

Selenium (Pytest):

POM, Parametrization, Exception Handling, Robust Locators.

BDD (Behave + Selenium):

Gherkin Scenarios, Scenario Outline, Examples Table, Data Tables (Filters, Products, Payments), Parametrization, Exception Handling, Robust Locators

Test Coverage:

End-to-End | Positive | Negative | Edge

Tools and Framework:

Selenium, Pytest, Allure Reporting

Utilities & Data:

Excel and CSV(Data-Driven), Logger, Screenshots, Config

HAIR MODULE:

This module automates the complete shampoo shopping flow on the Nykaa application. It validates key functionalities such as category navigation, product filters, product selection, cart management, and checkout process. The framework ensures end-to-end user journey testing using Selenium with POM and Pytest.

End-to-End Test Case

1. Launch the Nykaa website.
2. Navigate to the Hair Module from the homepage.
3. Selects the Shampoo Section.
4. Apply product filters such as Discount.
6. Selects the first featured product from product list.
7. Redirect to the product details page.
8. Validate product information and availability.
9. Click on Add to Bag.
10. Open the shopping cart/bag section.
11. Validate added product in the cart.
12. Click on Proceed to continue checkout flow.
13. Handles Checkout as Guest Popup.
14. Fetch data from CSV File to fill Address Section.
16. Click on Ship to this Address Button to continue.
17. Verify successful navigation towards checkout/payment page.

Positive Test Cases

- Add Product to Cart Test
- Cart Page Access Test
- Guest Checkout Flow Test
- Address Form Submission Test

Negative Test Cases

- Invalid Pincode Validation Test
- Empty Address Field Validation Test

MEN MODULE:

This module focuses on the Men category of the application. It includes testing functionalities like product navigation, filtering, sorting, and validation. The module ensures smooth user flow and proper integration with the overall automation framework.

Positive Test Cases

- Navigation Flow → Home → Payment → Pass
- Apply Filter (Valid Brand: Dove / Gucci) → Pass
- Search (Valid Input: Serums) → Pass
- Sorting Functionality → Pass
- Add to Cart → Cart Verification → Pass
- Checkout Navigation → Pass
- Empty Cart Validation → Pass
- Shipping Details (Valid Flow) → Pass

Negative Test Cases

- Invalid Filter (InvalidBrand) → No results → Pass
- Invalid Pincode → Error message → Pass
- Invalid Search (@@@@ / @\$%^) → Error handled → Pass
- Shipping Details (Invalid input) → Error message → Pass
- OTP Login Flow → Fail (Automation restriction)

End-to-End Test Case

Full Flow: Home → Men Page → Category page → Apply Filter and Sorting → Select Product → Add to Cart → Click cart and view details → Click Proceed → Click Continue as guest → Enter Shipping Details → Verifying Pincode → Payment Details → Checkout

Expected: Complete user journey without interruption

Result: Pass

FRAGRANCES MODULE:

This module automates the complete fragrance shopping flow on the Nykaa application. It validates key functionalities such as search, filter, product selection, cart management, and checkout process. The framework ensures end-to-end user journey testing using Selenium with POM and Pytest.

Positive Test Cases

- Search fragrance with valid keyword
- Apply filters (Brand, Price, Offers, Rating)
- Verify sorting options (Low to High, High to Low, Popularity)
- Validate product details page
- Add to bag and verify
- Update / remove items in bag
- Checkout with valid details
- Place order successfully
- Validate order confirmation

Negative Test Cases

- Search with invalid / special characters
- Invalid filter combinations
- Select out-of-stock product
- Add quantity beyond stock limit
- Incomplete delivery details
- Invalid mobile number / pincode
- No payment option selected
- Checkout failure scenarios

End-to-End Test Case

Launch application → Login popup handled → Navigate to Fragrance module

Search + Apply filters + Sort products

Select product → Validate details → Add to bag → Modify cart → Proceed to checkout → Enter delivery details

Select payment → Place order → Validate order confirmation

Logout / Close application

MAKEUP MODULE:

This module automates the end-to-end Lipstick shopping flow on Nykaa using Selenium WebDriver with Python. It validates core functionalities such as navigation, product listing, filtering, search, cart, and checkout. Framework follows POM design pattern with Pytest & BDD support..

Positive Test Cases

- TC01: Lipstick category page load
- TC02: Product display (name, price)
- TC03: Filter & sort functionality
- TC04: Valid product search

BDD Coverage

- Lipstick page navigation scenario
- Product visibility validation
- Filter & sort scenario
- Search functionality scenario
- Negative scenarios (invalid search, invalid URL)

Negative Test Cases

- TC05: Invalid search validation
- TC06: Invalid product URL handling

Key Areas Covered

- Homepage validation
- Makeup → Lipstick navigation
- Product listing & details validation
- Search & filter validation
- Add to Bag & Cart flow
- Checkout / Payment page validation

SKIN MODULE:

This module automates the complete moisturizer shopping flow on the Nykaa application. It validates key functionalities such as category navigation, product filtering, product selection, cart operations, and checkout process. The framework ensures end-to-end user journey testing using Selenium with POM and Pytest.

End-to-End Test Case

1. Launch the Nykaa website.
2. Navigate to the Skin Module from the homepage.
3. Open the Moisturizer sub-module/category.
4. Apply product filters such as brand selection.
5. Select the brand CeraVe.
6. Choose the first product from the filtered product list.
7. Redirect to the product details page.
8. Validate product information and availability.
9. Click on Add to Bag.
10. Open the shopping cart/bag section.
11. Validate added product in the cart.
12. Click on Proceed to continue checkout flow.
13. Verify successful navigation towards checkout/payment process.

Positive Test Cases

1. Cart Persistence After Page Refresh Test
2. Multiple Product Cart Validation Test
3. Remove Product from Cart Test
4. Search Bar Functionality Test

Negative Test Cases

1. Invalid Product Search Test
2. Proceed Button Visibility Validation for Empty Cart

HEALTH AND WELLNESS MODULE:

This module automates the complete Health & Wellness shopping flow on the Nykaa application. It validates key functionalities such as product search, filter validation, product selection, cart operations, and checkout process. The framework ensures end-to-end user journey testing using Selenium with POM and Pytest.

End-to-End Flow

Launch application → Navigate to Health & Wellness module → Search product → Validate listing → Apply filters & sorting → Select product → Validate details → Add to bag → Open cart → Update quantity → Proceed to checkout → Verify payment/login page

BDD Feature Scenarios

- Health & Wellness Module Navigation
- Product Search Functionality
- Product Display Validation
- Filter & Sort Functionality
- Add to Bag Functionality
- Cart Validation
- Invalid Search (Negative)
- Invalid Product URL (Negative)

Positive Test Cases

- Module Navigation Validation
- Product Search Functionality
- Product Display Validation
- Filter & Sort Functionality
- Product Detail Page Validation
- Add to Bag Functionality
- Cart Validation
- Proceed to Buy Validation

Negative Test Cases

- Invalid Product Search
- Empty Search Validation
- Invalid Quantity Selection
- Invalid Product URL

MOM AND BABY MODULE:

This module focuses on testing the Baby Care category in the Nykaa application. It validates functionalities like navigation, filtering, product selection, and cart flow using BDD scenarios. The automation framework is implemented using Selenium, Pytest, and POM design pattern.

End-to-End Flow

- Launch application → Navigate to Mom & Baby module
- Open Baby Care category
- Apply Brand + Discount filters
- Select product → Add to Bag
- Open Cart → Proceed to Checkout
- Continue as Guest → Enter shipping details
- Validate navigation to Payment Gateway
- Data-driven execution using CSV

BDD Feature Scenarios

- Category navigation validation
- Product display verification
- Brand filter functionality
- Search functionality
- Invalid filter handling
- End-to-end checkout flow
- Payment page validation

Positive Test Cases

Navigate to Baby Care category → Page load validation
Verify product display (names & prices)
Apply Brand filter (Mamaearth) → Products displayed
Apply Discount filter (10% and above) → Products displayed
Product selection → Product details validation
Add to Bag → Cart validation
Proceed to Buy → Payment/Login page reached

Negative Test Cases

Invalid brand search → No results displayed
Invalid discount (150%) → Option not available
Filter error handling → Page should not crash
Invalid input scenarios handled properly

THANK YOU